

# Adeel Shahbaz

AI & NLP Engineer | Computer Vision | Generative AI | Roman Urdu NLP Pioneer

Rawalpindi, Pakistan | National Skills University, Islamabad | AXI Technologies Trainee

## PROFILE

Computer Science student specializing in AI at National Skills University, Islamabad. Pioneer in Roman Urdu NLP — independently built the first-of-its-kind NLP toolkit and processing pipeline for Roman Urdu, a language with no prior standardized tooling. Skilled across Computer Vision, Generative AI, and autonomous systems. Currently training at AXI Technologies and AI Skill Bridge (NAVTTTC).

## EDUCATION

### Bachelor's in Computer Science — Specialization: Artificial Intelligence

National Skills University, Islamabad, Pakistan | Sep 2023 – May 2026 (Ongoing)

### NAVTTTC — AI (ML, DL & Communication)

AI Skill Bridge, Islamabad | Ongoing

## TECHNICAL SKILLS

|               |                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------|
| Languages     | Python, C++ (Android deployment via Python-to-Kotlin port)                                         |
| AI / ML       | PyTorch, ONNX Runtime, scikit-learn, HuggingFace Transformers, Generative AI                       |
| NLP           | Roman Urdu NLP Toolkit (first-of-its-kind), fastText, Custom Regex NER, Levenshtein Fuzzy Matching |
| Vision        | YOLOv8 / Ultralytics, OpenCV, Kalman Filtering                                                     |
| Robotics & HW | ROS, ArduPilot, Gazebo, MAVLink 2.0, Pixhawk 6C, Raspberry Pi, Ubuntu, 3D Printing (basic)         |
| Soft Skills   | Problem-solving, Research, Creativity, Team Collaboration                                          |

## PROJECTS

### KhudiChat — First Roman Urdu NLP Toolkit & Event Detection Pipeline

Role: NLP Architect (Roman Urdu Language Engine) | AI Skill Bridge / NAVTTTC

- Built the first Roman Urdu NLP toolkit — a groundbreaking contribution to a language that had no standardized NLP tooling, covering tokenization, entity extraction, fuzzy normalization, and event detection.
- Built a dual-routing pipeline: fastText detects language (English / Urdu / Roman Urdu) and routes to the appropriate engine — custom Roman Urdu NER for Urdu/Mixed, integrating with the team's MobileBERT (ONNX Runtime) gate for English.
- Ported the full Python NER pipeline (language detector + event extractor + pipeline orchestrator) to Kotlin, enabling fully offline on-device inference — no internet or server required.

### Autonomous Interceptor Drone — Vision-Based Rogue Drone Defense System

Role: AI Vision & Autonomy Engineer | AI Skill Bridge / NAVTTTC

- Co-developed a dual-layer autonomous interception system (Ground BCU + Aerial Drone) using YOLOv8 and Kalman Filtering — fully GPS-free.
- Hardware stack: Raspberry Pi (Ubuntu), Pixhawk 6C, 3D-printed drone frame; validated in Gazebo & SITL simulation.
- Achieved precision engagement within 3 metres at ~\$628 — approximately 56x cheaper than military alternatives.

## WORK EXPERIENCE

### AXI Technologies

Trainee | AI / Machine Learning | Rawalpindi, Pakistan

- Hands-on training in AI/ML model development and real-world deployment pipelines covering computer vision and generative AI.

### AI Skill Bridge

Trainee | NAVTTTC — AI (ML, DL & Communication)

- Delivered two end-to-end AI projects; pioneered the Roman Urdu NLP toolkit as a standout original contribution within the programme.